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Sending Large Declarations

AS4-Gateway

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Sending Large Declarations – AS4-Gateway

Document change history

Version	Description	Who	When
1.0	First version	Ulrik Andersen	20-01-2026
1.1	Following sections modified after review: • Introduction • Limitations (added table with a summary of use cases) • Further recommendations (added link)	Ulrik Andersen	23-01-2026
1.1.1	Minor corrections to the section Introduction.	Ulrik Andersen	23-01-2026

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Introduction

All messages to the AS4-Gateway are signed; the process of signing can be resource-intensive for very large documents, for both the sender and the receiver of an AS4 message. To address this, then it is possible to compress a document before sending them with an AS4 message.

There are the following benefits of using compression:

- Less traffic to send over network. File size can be reduced to a factor of 20 to 25 (average) depending on the content and structure of the XML document.
- When signing a non-compressed XML document, the XML document is canonicalized using the Exclusive XML Canonicalization algorithm. This process, along with the signature generation, is quite CPU-intensive for large XML documents. Thus, it is much faster to simply sign the (smaller) compressed file.

When to compress a document is a trade-off between the time it takes to compress a document and the time it takes to canonicalize and sign an XML document, and it can be difficult to give a precise rule to when the first is faster than the latter.

However, the AS4-Gateway has a limit on the size of an AS4 message of 20 MB, so if a document is larger than this, then it must be compressed.

Requirements

When using software that supports the AS4 profile, it is quite simple to send a compressed declaration document in an AS4 message. For more information please refer to the specification: [3.1 Compression - AS4 Profile of ebMS 3.0 Version 1.0]

See further sub-sections for details on how to send a compressed file.

Compress file

Compress the declaration document using gzip (it MUST be gzip and not any other compression algorithm).

AS4 user message settings

Include the compressed file as an attachment to the AS4 message in the same way as an uncompressed file is sent, using the following settings in the AS4 user message:

```
<PartInfo href="…">
  <PartProperties>
    <Property name="MimeType">application/xml</Property>
    <Property name="CompressionType">application/gzip</Property>
    <Property name="CharacterSet">utf-8</Property>
  </PartProperties>
</PartInfo>
```

Some notes:

- The MimeType property contains the mime type of the zipped file; in the case of declarations this will always be application/xml.
- The CompressionType property must be set to application/gzip.
- The CharacterSet property is optional, but if using another character encoding than UTF-8, then it is strongly recommended to include information about the character encoding.

Limitations

There will be a limit on the file size of an AS4 message of 20MB. This means that a declaration document must be compressed, if it exceeds this limit.

The limit of 20MB also applies to bulk upload. So, when using bulk upload, the total size of the uploaded declaration documents, must not exceed 20MB.

Overview of size limits for different use cases.

Sending	Size limt
Uncompressed declaration	The size of the declaration is limited to 20 MB
Compressed declaration	The size of the compressed declaration is limited to 20 MB(*).
Bulk upload – uncompressed declarations	The total size of the attached declarations must not exceed 20 MB.
Bulk upload – compressed declarations	The total size of the attached compressed declarations must not exceed 20 MB(*).
Bulk upload – mix of compressed and uncompressed declarations	The total size of the attached declarations (compressed or not compressed) must not exceed 20 MB(*).

(*) Currently DMS has a size limit for a declaration of 24 MB. So, a single uncompressed declaration must not exceed 24 MB.

Further recommendations

It is recommended to use the bulk upload endpoint for sending large files even if only 1 compressed file is sent.

The reason for this is that this endpoint returns a receipt as soon as the AS4 message is received and verified, whereas the “standard” endpoint does not return a response before the payload is validated, which in case of larger documents will take longer time. Thus, using the “bulk upload” endpoint will reduce the response time and thereby improve the throughput.

For more information on bulk upload please refer to this document: AS4 Gateway - multiple documents in a request.pdf

Relevant links

- Compression in the AS4 profile: https://docs.oasis-open.org/ebxml-msg/ebms/v3.0/profiles/AS4-profile/v1.0/os/AS4-profile-v1.0-os.html#__RefHeading__26170_1909778835
- SOAP Message Security 1.1: https://docs.oasis-open.org/wss/v1.1/wss-v1.1-spec-errata-os-SOAPMessageSecurity.pdf
- SOAP Messages with Attachments: https://www.w3.org/TR/SOAP-attachments/
- Web Services Security SOAP Message with Attachments (SwA) Profile Version 1.1.1: https://docs.oasis-open.org/wss-m/wss/v1.1.1/cos01/wss-SwAProfile-v1.1.1-cos01.html
- Exclusive XML Canonicalization Version 1.0: https://www.w3.org/TR/2002/REC-xml-exc-c14n-20020718/